

POULTRYWISE

Broiler Farming Masterclass

A Step-by-Step Guide to High-Yield Commercial Meat Production

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A Professional Guide for Modern Small and Commercial Poultry Farmers

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About Book 3

This masterclass provides comprehensive guidelines on raising fast-growing broiler chickens efficiently, covering housing, nutrition, health, and marketing to achieve maximum return on investment.

Chapter 1: Overview of Commercial Broiler Production

Commercial broiler farming focuses on raising fast-growing chickens specifically for meat production. The primary operational objective is achieving maximum final body weight within a short cycle (typically 35 to 42 days) while maintaining a low Feed Conversion Ratio (FCR).

Key Operational Advantages:

- **Fast Turnaround:** Broilers reach market weight in 5 to 6 weeks, allowing 5–6 grow-out cycles per year.
- **Predictable Growth:** Modern genetic strains offer predictable growth rates when target environment and nutrition are met.
- **High Protein Demand:** Meat production caters to steady and growing market demand.

Chapter 2: Selecting High-Performing Broiler Strains

Choosing the right commercial strain depends on market demands, target processing weight, and local environmental conditions.

• Cobb 500

- Yield & FCR Efficiency
- Rapid early growth
- World's popular broiler; exceptional feed conversion and uniform growth.

• Ross 308

- Overall Robustness
- Balanced growth curve
- Excellent yield, strong legs, and adaptable to diverse climate conditions.

• Hubbard Classic

- Adaptability
- Steady, resilient growth
- Performs well under variable management environments and tropical heat.

• Arbor Acres Plus

- Breast Meat Yield
- Excellent body conformation
- Consistent growth rate and low overall mortality rates.

Strain	Primary Focus	Growth Profile	Key Characteristics
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Chapter 3: Housing Setup & Environmental Control

Proper housing protects birds from extreme weather, controls humidity, and ensures adequate airflow.

- **Open-Sided Housing:** Cost-effective in warm climates; requires side curtains and circulation fans to manage airflow.
- **Closed Environment-Controlled Housing:** Uses negative-pressure tunnel ventilation and evaporative cooling pads to strictly regulate temperature, humidity, and airflow.
- **Stocking Density:** Maintain 10 to 12 birds per square meter in open houses and up to 15 birds per square meter in environmentally controlled systems.

Chapter 4: Critical Brooding Phase (Days 1–14)

The first two weeks dictate lifetime performance. Early chick care establishes cardiovascular health, skeleton size, and gut development.

- **Pre-Heating:** Turn on heat sources 24–36 hours before chicks arrive to ensure floor/litter temperature reaches 32°C–34°C.
- **Crop Fill Assessment:** Check crop fill 8 and 24 hours post-placement. At 24 hours, at least 95% of chicks should have soft, full crops containing feed and water.
- **Brooding Temperature Schedule:**
 - Days 1–3: 32°C–34°C
 - Days 4–7: 30°C–31°C
 - Days 8–14: 27°C–29°C

Chapter 5: Broiler Nutrition & Feed Conversion Optimization

Feed accounts for 65%–70% of total broiler production costs. Formulating or feeding high-quality rations ensures maximum gain per kilogram of feed consumed.

Nutritional Stages:

- **Broiler Starter (Days 0–10):** High protein (22%–23% CP) in crumb format to support early frame and gut growth.
- **Broiler Grower (Days 11–24):** Balanced protein (20%–21% CP) and energy in mini-pellet/pellet form for muscle development.
- **Broiler Finisher (Days 25 to Market):** High energy, slightly lower protein (18%–19% CP) in pellet form to maximize final body mass.

FCR Benchmark:

Aim for an FCR of 1.50 to 1.65 (1.5 kg to 1.65 kg of feed per 1 kg of live weight gain).

Chapter 6: Water Quality & Litter Management

Clean water and dry litter are fundamental to broiler health and footpad skin integrity.

- **Water Intake:** Broilers drink approximately 1.8 to 2 times more water than the weight of feed consumed.
- **Water Sanitation:** Treat drinking water with chlorine or hydrogen peroxide solutions; maintain pH between 6.0 and 6.8.
- **Litter Care:** Keep litter depth at 5–8 cm using wood shavings or chopped straw. Stir wet patches regularly to prevent ammonia buildup and footpad dermatitis.

Chapter 7: Biosecurity, Vaccination & Health Management

Preventing disease outbreak is significantly cheaper than treating sick flocks.

- **Core Vaccinations:** Administer Marek's Disease (at hatchery), Newcastle Disease (ND), and Infectious Bursal Disease (Gumboro/IBD) based on local challenge levels.
- **Biosecurity Protocol:** Enforce mandatory footbaths, vehicle disinfection zones, and farm-specific footwear/apparel.
- **All-In / All-Out:** Clear entire houses simultaneously and maintain a strict 14-day downtime for washing, sanitizing, and resting houses between batches.

Chapter 8: Daily Monitoring & Growth Benchmarking

Regular monitoring enables rapid intervention when growth metrics deviate from target curves.

- **Weight Sampling:** Weigh 100 birds or 1% of the house population weekly to monitor Average Daily Gain (ADG) and flock uniformity.
- **Uniformity Goal:** At least 80%–85% of birds should fall within $\pm 10\%$ of the average flock weight.
- **Mortality Tracking:** Target total batch mortality of less than 3% to 4%.

Chapter 9: Catching, Transport & Processing Management

Improper harvesting can ruin weeks of diligent rearing through bruising, bone fractures, or weight loss due to stress.

- **Feed Withdrawal:** Remove feed 8 to 12 hours prior to slaughter to clear gut contents while preventing water loss and intestinal rupture during processing.
- **Catching Protocols:** Dim house lights or use blue light during catching to keep birds calm. Hold birds carefully by the body or legs to avoid injury.
- **Transport:** Schedule transport during cooler hours (night or early morning) to prevent heat stress and field mortality.

Chapter 10: Cost Efficiency & Financial Management

To ensure high profitability, measure key performance indicators (KPIs) for every finished batch.

European Production Efficiency Factor (EPEF):

$$EPEF = [(Survival\ Rate\ \% \times Average\ Live\ Weight\ kg) / (Age\ in\ Days \times FCR)] \times 100$$

Target EPEF: A score above 350 indicates excellent management efficiency.

- **Cost Controls:** Consistently analyze cost per kilogram of live weight produced across feed, day-old chicks, energy, and labor.

--- End of Book 3 (PoultryWise Broiler Farming Masterclass) ---